Supplementary Methods III

DNA extraction and genomic analyses of propionate oxidisers and butyrate oxidisers

Literature investigation identified three microorganisms as syntrophic propionate oxidisers (*Syntrophobacter wolinii*, *Syntrophobacter sulfatireducens* and *Syntrophobacter pfennigii*; [1–3]) and three as syntrophic butyrate oxidisers (*Syntrophomonas buswelli*, *Syntrophomonas erecta* and *Syntrophomonas curvata*; [4–6]) for which no genome sequences were publicly available (*Table III-S1*). Therefore, three of them were purchased as active cellular cultures and underwent DNA extraction, while for the other three the DNA was directly purchased. DNA extraction was performed according to the commercial kit's instructions, RiboPure-Yeast Invitrogen (Thermo Fisher Scientific, Waltham, MA, USA). For each extract, before and after DNase treatment, Nanodrop2000 (Thermo Fisher Scientific, Waltham, MA, USA) and Qubit fluorometers (Thermo Fisher Scientific, Waltham, MA, USA) were used to assess DNA quality and concentration, respectively. DNA of all six microorganisms was sent for sequencing to the DiBio Sequencing service (Department of Biology, University of Padova, Italy), where library preparation was performed using the Nextera DNA Flex Library Prep Kit (Illumina Inc, San Diego, CA) and the sequencing was performed with the Illumina NovaSeq platform (2 × 150, paired-end).

Bioinformatics analysis was performed to obtain the genomic sequences of each syntrophic propionate and butyrate oxidisers. Raw read quality was assessed using FastQC v0.11.8 [7] and reads were trimmed with trimmomatic software [8]. Reads were assembled into genome assemblies using SPAdes v3.13.0, contigs shorter than 1000 bp were removed and the assembly quality was assessed with QUAST v5.0.2 [9, 10]. Genome size and other parameters suggested the presence of contaminants, further confirmed by taxonomic analysis; therefore, we treated these as metagenomic samples and not as isolates, and we extracted the genomes using a binning strategy. For read alignment, the reference genome index was built with Bowtie2 v2.4.5 and contigs were binned into MAGs using MetaBAT2 v2.2.15 [11, 12]. MAGs' quality, completeness and contamination level were assessed using CheckM2 v0.1.3 (*Table III-S1*) [13]. Finally, taxonomic classification of the MAGs was performed using GTDB-Tk v2.3.2 [14]. The obtained sequences were added to genomes obtained from the literature to generate the corresponding positive classes of the function-specific training sets.

Negative classes refinement for construction of function-specific training sets

For functions sharing genomic components with related metabolic processes, genomes representing biologically relevant alternative functions were explicitly added to the negative class, so that the models were trained to distinguish the target functions from similar processes that could otherwise generate false-positive predictions. For the homoacetogenesis model, methanogens, sulfate reducers and syntrophic propionate- and butyrate-oxidising microorganisms were included as negatives because these organisms can encode components of the WLP that are also present in homoacetogens. For starch degradation, sulfate reducers and syntrophic propionate- and butyrate-oxidising microorganisms were included because these AD-associated groups were not represented in the original MICROPHERRET background [15] and could otherwise generate false-positives. Similarly, hydrogen-consuming methanogens, sulfate reducers and homoacetogens were included as negatives for H_2 production to provide biologically relevant H_2-consuming counterparts that were insufficiently represented in the original background. These additional genomes were incorporated after the KO-based refinement step described above.

References

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Supplementary Figures III

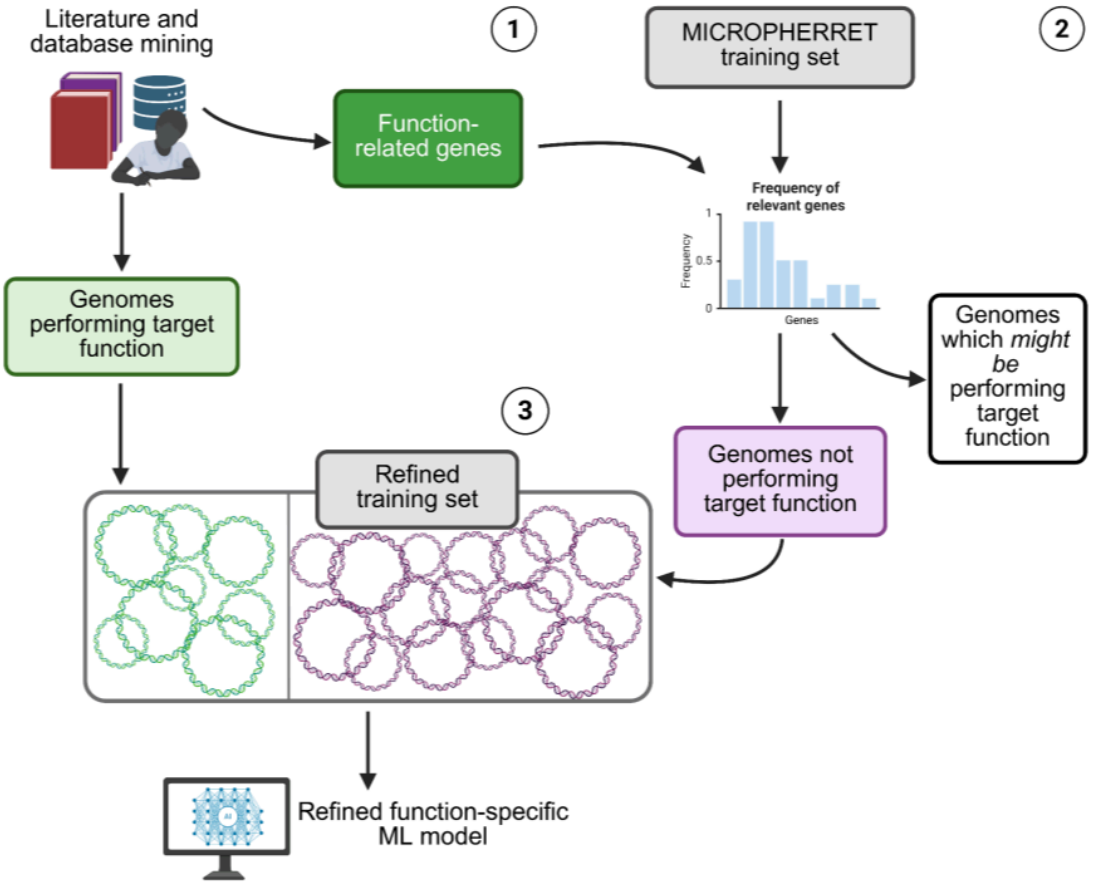


Figure III-S1. **Workflow used to generate each refined function-specific training set from the original MICROPHERRET dataset.** Literature and databases were scanned to retrieve genomes associated with each target function and the corresponding function-related genes. These genes were used to identify genomes in the MICROPHERRET training set that were unlikely to perform the target function. The newly identified genomes and literature-derived positive representatives were then used to construct the refined training set, forming its negative and positive classes.

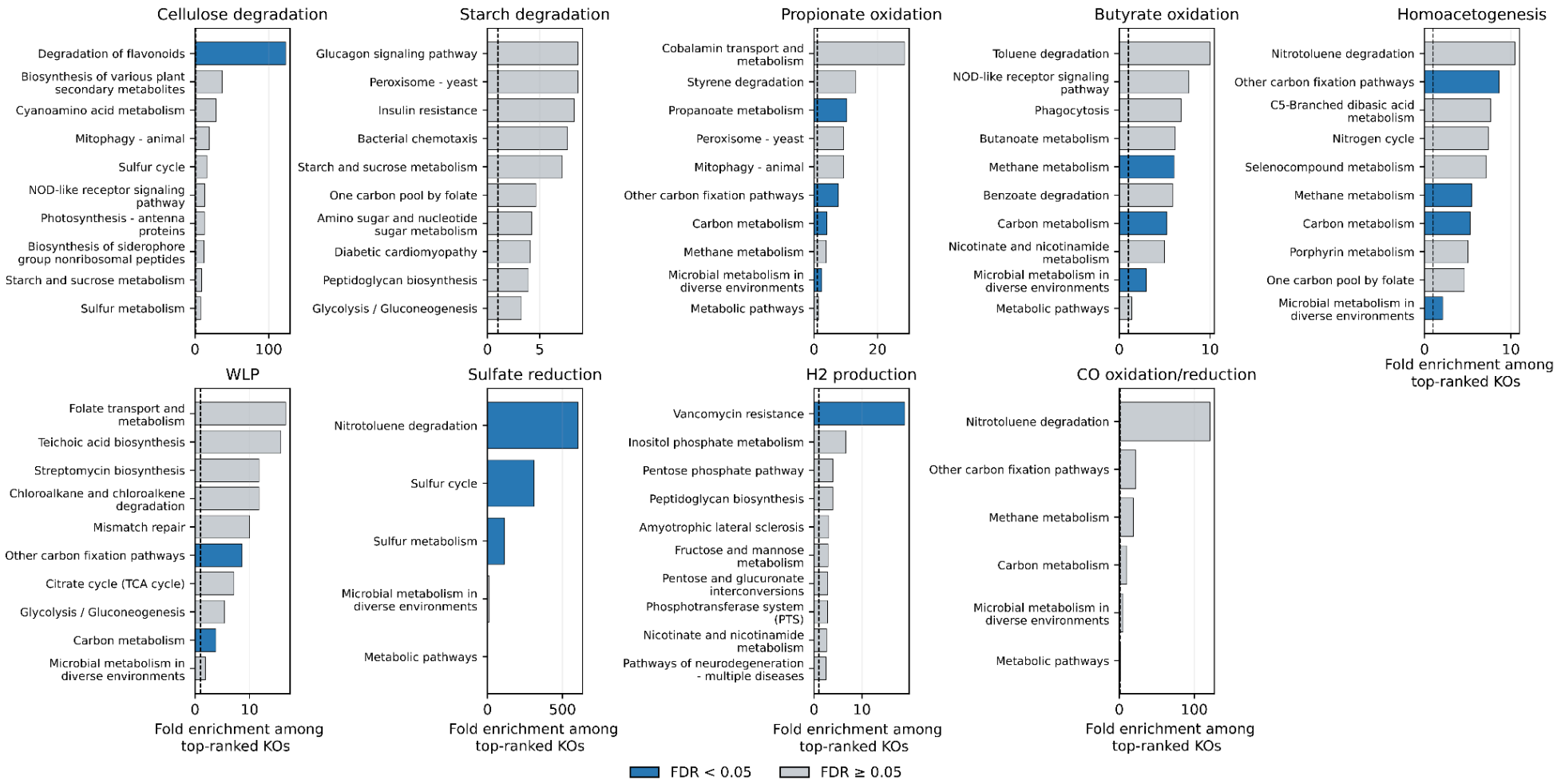

Cellulose degradation
Degradation of flavonoids
Biosynthesis of various plant secondary metabolites
Cyanoamino acid metabolism
Mitophagy - animal
Sulfur cycle
NOD-like receptor signaling pathway
Photosynthesis - antenna proteins
Biosynthesis of siderophore group nonribosomal peptides
Starch and sucrose metabolism
Sulfur metabolism
0
100
Starch degradation
Glucagon signaling pathway
Peroxisome - yeast
Insulin resistance
Bacterial chemotaxis
Starch and sucrose metabolism
One carbon pool by folate
Amino sugar and nucleotide sugar metabolism
Diabetic cardiomyopathy
Peptidoglycan biosynthesis
Glycolysis / Gluconeogenesis
0
5
Propionate oxidation
Cobalamin transport and metabolism
Styrene degradation
Propanoate metabolism
Peroxisome - yeast
Mitophagy - animal
Other carbon fixation pathways
Carbon metabolism
Methane metabolism
Microbial metabolism in diverse environments
Metabolic pathways
0
20
Butyrate oxidation
Toluene degradation
NOD-like receptor signaling pathway
Phagocytosis
Butanoate metabolism
Methane metabolism
Benzoate degradation
Carbon metabolism
Nicotinate and nicotinamide metabolism
Microbial metabolism in diverse environments
Metabolic pathways
0
10
Homoacetogenesis
Nitrotoluene degradation
Other carbon fixation pathways
C5-Branched dibasic acid metabolism
Nitrogen cycle
Selenocompound metabolism
Methane metabolism
Carbon metabolism
Porphyrin metabolism
One carbon pool by folate
Microbial metabolism in diverse environments
0
10
Fold enrichment among top-ranked KOs
WLP
Folate transport and metabolism
Teichoic acid biosynthesis
Streptomycin biosynthesis
Chloroalkane and chloroalkene degradation
Mismatch repair
Other carbon fixation pathways
Citrate cycle (TCA cycle)
Glycolysis / Gluconeogenesis
Carbon metabolism
Microbial metabolism in diverse environments
0
10
Fold enrichment among top-ranked KOs
Sulfate reduction
Nitrotoluene degradation
Sulfur cycle
Sulfur metabolism
Microbial metabolism in diverse environments
Metabolic pathways
0
500
Fold enrichment among top-ranked KOs
H2 production
Vancomycin resistance
Inositol phosphate metabolism
Pentose phosphate pathway
Peptidoglycan biosynthesis
Amyotrophic lateral sclerosis
Fructose and mannose metabolism
Pentose and glucuronate interconversions
Phosphotransferase system (PTS)
Nicotinate and nicotinamide metabolism
Pathways of neurodegeneration - multiple diseases
0
10
Fold enrichment among top-ranked KOs
CO oxidation/reduction
Nitrotoluene degradation
Other carbon fixation pathways
Methane metabolism
Carbon metabolism
Microbial metabolism in diverse environments
Metabolic pathways
0
100
Fold enrichment among top-ranked KOs
FDR < 0.05
FDR ≥ 0.05

Figure III-S2. **KEGG pathway enrichment among the most informative KOs for the nine predicted functions.** Each panel represents one function and shows up to ten pathways with the lowest FDR among those containing at least two KOs within the top 1% of the corresponding model features. Features were ranked using the model-specific importance measure, and enrichment was calculated relative to all candidate KOs available to the corresponding model. Bar length indicates fold enrichment, while the dashed vertical line indicates no enrichment. Blue bars indicate significant enrichment following Fisher's exact test and Benjamini–Hochberg correction within each function (FDR < 0.05); grey bars indicate non-significant enrichment.

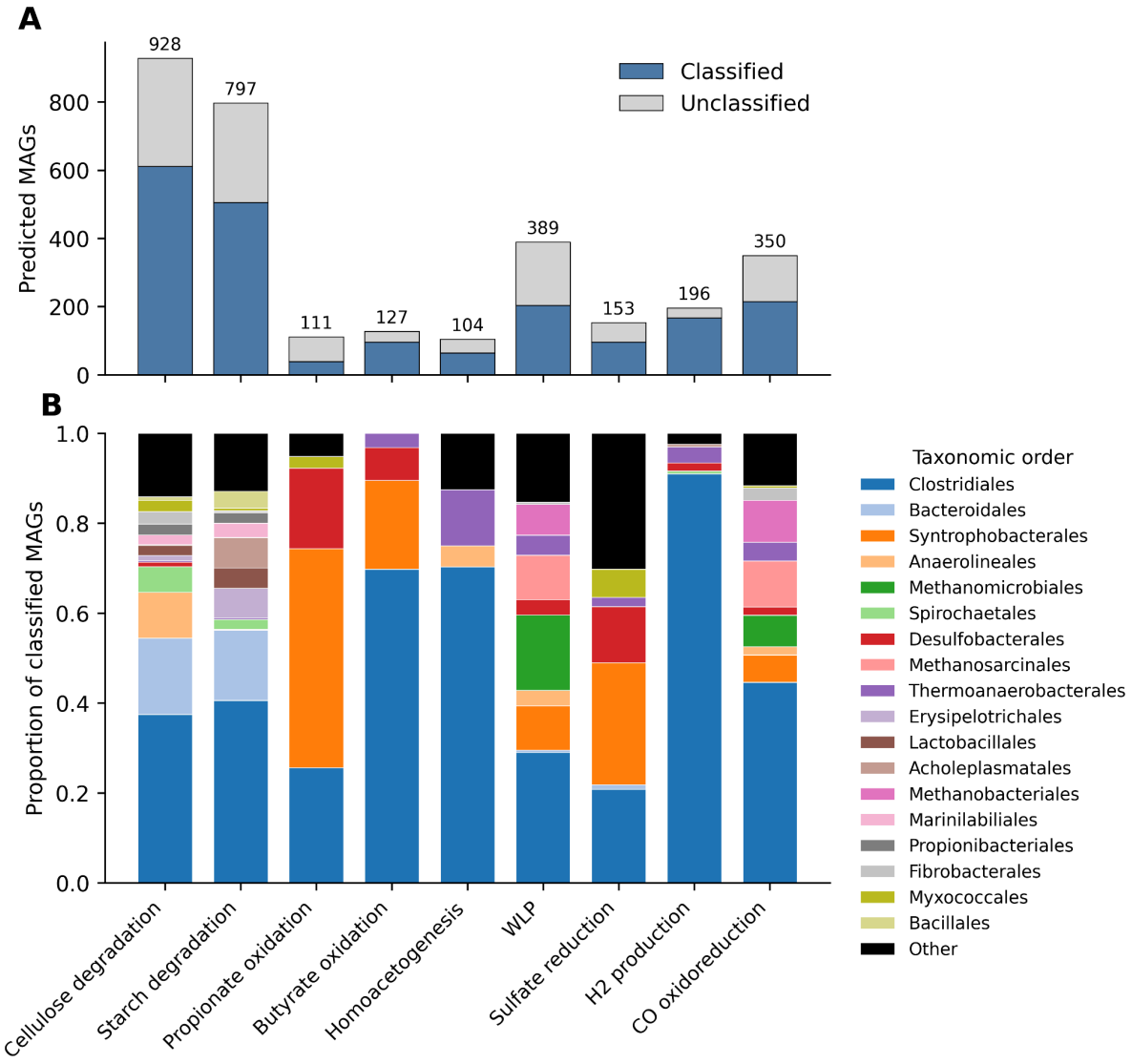


Figure III-S3. **Number and taxonomic composition of MAGs predicted to perform the nine AD functions in the Biogas Microbiome database.** A) Number of predicted MAGs with or without a taxonomic classification at the order level. Numbers above the bars indicate the total number of MAGs predicted for each function. B) Taxonomic order of the classified MAGs. Unclassified MAGs were excluded and each bar was normalised to one to allow comparison among functions with different numbers of predictions. Colours represent the 18 most abundant orders across the nine functions, while the remaining were grouped as Other.

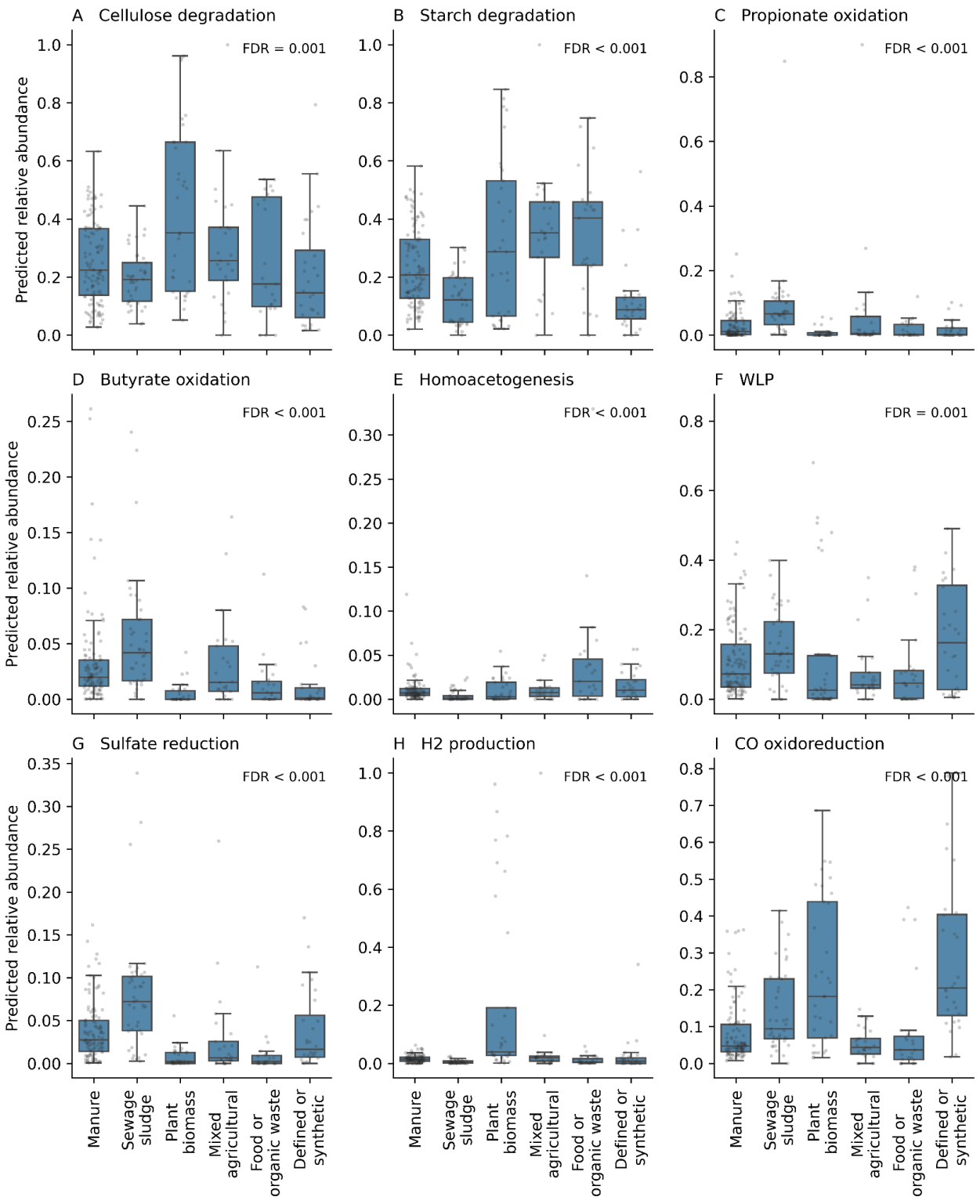


Figure III-S4. **Predicted abundance of the nine functional populations across feedstock groups.** Each panel shows the summed relative abundance of MAGs predicted to perform one function across samples grouped by Feedstock_source. Points represent individual samples and boxplots summarise their distributions. The FDR reported above each panel was obtained by applying the Benjamini-Hochberg correction to the Kruskal-Wallis p-values calculated for the nine functions.

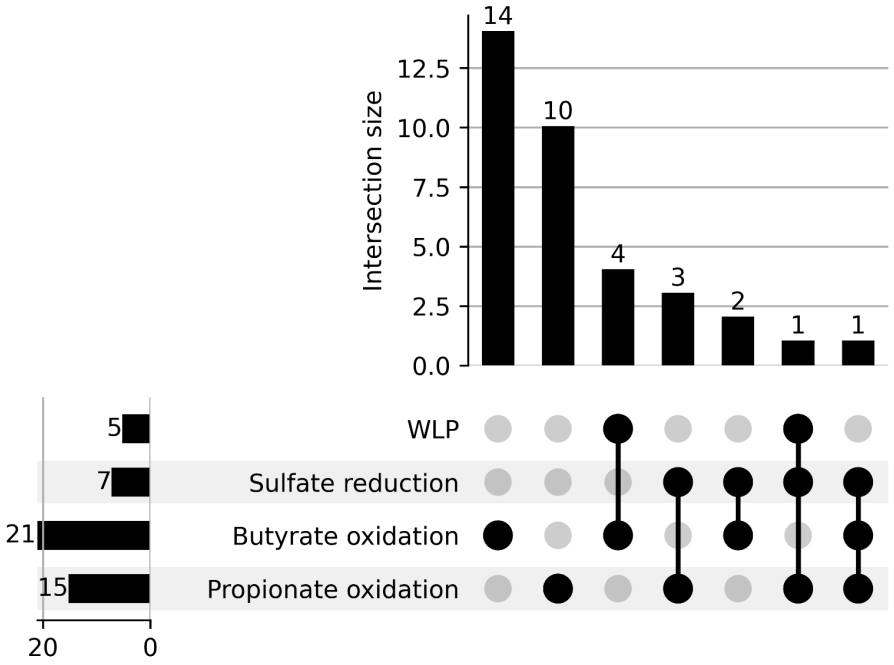


Figure III-S5. **Overlap of functional predictions among propionate- and butyrate-oxidising MAGs in the samples from Hao *et al***. The plot shows the intersections among propionate oxidation, butyrate oxidation, sulfate reduction and the Wood–Ljungdahl pathway. Vertical bars report the number of MAGs in each observed combination, while connected points indicate the functions included in that intersection. Horizontal bars report the total number of MAGs assigned to each function.

Supplementary Tables III

Table III-S1. **Positive reference genomes used to construct the function-specific training datasets**. Genomes are reported separately for each of the nine functions together with their species name, taxonomic order, NCBI RefSeq accession, functional role and links to the available NCBI records. Assembly level is reported for publicly available genomes, while completeness and contamination are additionally provided for the genomes sequenced and reconstructed in this study. Genomes assigned to more than one function are reported separately within each corresponding functional class.

Table III-S2. **Training-set composition, model selection, hyperparameters, cross-validation results and KO features for the nine AD functions**. For each function, the table reports the number of positive and negative training genomes, the selected algorithm and its hyperparameters, and the complete cross-validation summary for the evaluated LR, RF and SVM configurations. The cross-validation information includes the fold-specific selected hyperparameters and training and validation F1 and MCC values. The number of KOs retained as important by the final fitted model and the first 100 KOs ranked by coefficient magnitude or feature importance are also reported.

Table III-S3. **Metadata and feedstock classification of samples used from the Biogas Microbiome database.** Sample identifier, reactor configuration, scale, operating temperature, influent feedstock and main feedstock composition were obtained from the original database. Information on the influent substrates was used in this study to assign each sample to one Feedstock_source group and to generate non-mutually exclusive Boolean indicators describing the presence of specific substrate components (e.g., carbohydrates, proteins, lipids).

Table III-S4. **Summary of the datasets used as AD case studies.** For each study, the analysed samples, reactor configuration, experimental conditions and comparisons addressed in the corresponding analysis are reported. The table also describes the operational phases or treatments used to interpret variation in the predicted functional populations.